

Python Complete Course (Basic to Advanced)

Module 1: Introduction to Python

Lesson 1

- What is Python?
- History of Python
- Features of Python
- Applications of Python
- Python Versions (Python 2 vs Python 3)
- Why Python?
- Installing Python
- Installing VS Code
- Installing PyCharm
- Running Python Programs
- Interactive Mode (REPL)
- Script Mode

Commands

```
python --version  
python3 --version  
pip --version
```

First Program

```
print("Hello, World!")
```

Module 2: Python Basics

Lesson 2

- Python Syntax
- Comments
- Indentation

- Variables
- Naming Rules
- Keywords
- Identifiers

Practice

```
name = "John"  
age = 25  
salary = 50000
```

Module 3: Data Types

Lesson 3

Built-in Data Types

- int
- float
- complex
- bool
- str
- list
- tuple
- set
- dict
- bytes
- bytearray
- range
- NoneType

Type Checking

```
type()  
isinstance()
```

Projects

- Student Information
-

Module 4: Operators

Lesson 4

- Arithmetic Operators
- Assignment Operators
- Comparison Operators
- Logical Operators
- Bitwise Operators
- Membership Operators
- Identity Operators

Projects

- Calculator
-

Module 5: Input & Output

Lesson 5

- input()
- print()
- Escape Characters
- String Formatting
- f-Strings

Projects

- Student Marks Calculator
-

Module 6: Conditional Statements

Lesson 6

- if
- if-else
- elif
- Nested if
- Match Case (Python 3.10+)

Projects

- Grade Calculator
 - ATM Menu
-

Module 7: Loops

Lesson 7

- for Loop
- while Loop
- Nested Loops
- break
- continue
- pass

Projects

- Multiplication Table
 - Prime Numbers
 - Pattern Printing
-

Module 8: Strings

Lesson 8

- String Indexing
- Slicing
- String Methods (50+)
- String Formatting
- Regular Expressions (Introduction)

Projects

- Password Validator
 - Email Validator
-

Module 9: Lists

Lesson 9

- Creating Lists
- List Methods
- Nested Lists
- List Comprehensions

Projects

- Shopping Cart
 - Student Marks List
-

Module 10: Tuples

Lesson 10

- Tuple Operations
- Packing & Unpacking
- Tuple Methods

Projects

- Employee Records
-

Module 11: Sets

Lesson 11

- Set Operations
- Union
- Intersection
- Difference
- Symmetric Difference

Projects

- Student Attendance
-

Module 12: Dictionaries

Lesson 12

- Creating Dictionaries
- Dictionary Methods
- Nested Dictionaries
- Dictionary Comprehensions

Projects

- Employee Database
-

Module 13: Functions

Lesson 13

- Function Definition
- Arguments
- Keyword Arguments
- Default Arguments
- Variable-Length Arguments
- Lambda Functions
- Recursion

Projects

- EMI Calculator
 - Factorial Using Recursion
-

Module 14: Modules & Packages

Lesson 14

- import
- from...import
- Creating Modules

- Packages
- pip

Projects

- Utility Package
-

Module 15: File Handling

Lesson 15

- Reading Files
- Writing Files
- CSV Files
- JSON Files
- Binary Files

Projects

- Student File Manager
-

Module 16: Exception Handling

Lesson 16

- try
- except
- else
- finally
- raise
- Custom Exceptions

Projects

- Banking System
-

Module 17: Object-Oriented Programming (OOP)

Lesson 17

- Class
- Object
- Constructor
- Instance Variables
- Class Variables
- Instance Methods
- Static Methods
- Class Methods

Projects

- Student Management System
-

Module 18: Advanced OOP

Lesson 18

- Inheritance
- Polymorphism
- Encapsulation
- Abstraction
- Method Overloading
- Method Overriding
- Operator Overloading

Projects

- Banking Application
-

Module 19: Advanced Python

Lesson 19

- Iterators
- Generators
- Decorators
- Closures
- Namespaces
- Scope (LEGB)

Projects

- Custom Decorators
-

Module 20: Regular Expressions

Lesson 20

- Pattern Matching
- Search
- Match
- Findall
- Split
- Sub

Projects

- Email Validator
 - Phone Number Validator
-

Module 21: Date & Time

Lesson 21

- datetime
- calendar
- time

Projects

- Age Calculator
- Attendance Tracker

Module 22: Database Programming

Lesson 22

- SQLite
- MySQL Connector
- CRUD Operations
- Transactions

Projects

- Student Database
 - Employee Database
-

Module 23: GUI Development

Lesson 23

- Tkinter
- Labels
- Buttons
- Frames
- Entry
- Menus
- Dialog Boxes

Projects

- Calculator
 - Notepad
-

Module 24: Multithreading & Multiprocessing

Lesson 24

- Threads
- Processes
- Synchronization
- ThreadPool

Projects

- File Downloader
-

Module 25: Networking

Lesson 25

- Socket Programming
- Client-Server Communication
- HTTP Requests

Projects

- Chat Application
-

Module 26: Web Scraping & APIs

Lesson 26

Libraries

- requests
- BeautifulSoup
- Selenium (Introduction)

Projects

- News Scraper
 - Weather API
-

Module 27: Testing & Logging

Lesson 27

- unittest
- pytest (Introduction)
- Logging
- Debugging

Projects

- Student System Testing
-

Module 28: Python Automation

Lesson 28

Libraries

- os
- shutil
- smtplib
- schedule

Projects

- File Organizer
 - Email Sender
 - Folder Backup
-

Module 29: Performance & Best Practices

Lesson 29

- Virtual Environments

- Code Optimization
 - PEP 8
 - Type Hints
 - Profiling
 - Packaging
-

Module 30: Final Projects

Build Complete Real-World Applications

Projects

1. Student Management System
 2. Employee Management System
 3. Library Management System
 4. Hospital Management System
 5. Banking System
 6. Inventory Management System
 7. Hotel Management System
 8. School ERP
 9. Online Examination System
 10. Learning Management System (LMS)
 11. ATM Simulation
 12. Billing Software
 13. Expense Tracker
 14. Contact Management System
 15. Quiz Application
 16. Weather App
 17. News Scraper
 18. Chat Application
 19. File Encryption Tool
 20. Face Recognition Attendance System
 21. Voice Assistant
 22. QR Code Generator
 23. PDF Generator
 24. Password Manager
 25. URL Shortener
 26. Web Crawler
 27. Image Processing Tool
 28. Email Automation System
 29. REST API Client
 30. AI-Powered Python Application
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30 Hands-on Practice Programs

1. Hello World
2. Variables & Data Types
3. Calculator
4. Even/Odd Checker
5. Prime Number
6. Armstrong Number
7. Fibonacci Series
8. Pattern Printing
9. String Reversal
10. Palindrome Checker
11. List Operations
12. Tuple Operations
13. Dictionary CRUD
14. Set Operations
15. Function Calculator
16. Lambda Examples
17. File Reader
18. CSV Processor
19. JSON Reader
20. Student Database
21. Banking System
22. Employee Payroll
23. ATM System
24. Tkinter Calculator
25. Socket Chat
26. Weather API
27. File Organizer
28. Email Automation
29. Student Management System
30. Complete ERP Project

Python Interview Questions (Basic to Advanced)

Beginner

1. What is Python?
2. What are Python's features?

3. What are Python data types?
4. What is the difference between a list and a tuple?
5. What are mutable and immutable objects?
6. Explain Python indentation.
7. What is `None`?
8. Difference between `==` and `is`?
9. What are Python keywords?
10. What are Python modules?

Intermediate

11. What are functions?
12. Explain lambda functions.
13. What is recursion?
14. Explain OOP concepts in Python.
15. Difference between `append()` and `extend()`?
16. What are generators?
17. What are decorators?
18. Explain exception handling.
19. What is file handling?
20. Explain list comprehensions.

Advanced

21. Explain the GIL (Global Interpreter Lock).
 22. What are iterators?
 23. Difference between multithreading and multiprocessing?
 24. Explain context managers (`with` statement).
 25. What is monkey patching?
 26. Explain metaclasses (Introduction).
 27. How does Python memory management work?
 28. Explain shallow copy vs deep copy.
 29. What are virtual environments?
 30. What are Python best practices?
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20 Real-World Python Projects

1. Student Management System
2. Employee Management System
3. Banking System
4. Library Management System
5. Hospital Management System
6. School ERP
7. Hotel Management System

8. Inventory Management System
 9. Billing Software
 10. Expense Tracker
 11. Weather Application
 12. News Scraper
 13. Chat Application
 14. Face Recognition Attendance System
 15. QR Code Generator
 16. Password Manager
 17. Email Automation Tool
 18. Voice Assistant
 19. AI Chatbot
 20. Learning Management System
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Python Cheat Sheet

Variables

```
name = "John"  
age = 25
```

Input & Output

```
name = input("Enter Name: ")  
print(f"Hello {name}")
```

Loops

```
for i in range(5):  
    print(i)
```

```
while True:  
    break
```

Functions

```
def add(a, b):  
    return a + b
```

File Handling

```
with open("file.txt", "r") as file:  
    data = file.read()
```

OOP

```
class Student:  
    def __init__(self, name):
```



```
self.name = name
```

Exception Handling

```
try:  
    x = 10 / 0  
except ZeroDivisionError:  
    print("Cannot divide by zero")
```

Database

```
import sqlite3  
conn = sqlite3.connect("students.db")
```

Virtual Environment

```
python -m venv venv  
source venv/bin/activate # macOS/Linux  
venv\Scripts\activate    # Windows
```

Package Installation

```
pip install requests  
pip install numpy  
pip install pandas
```